

Module 4 (Machine Shop)

Principle of metal removal operation, Machine tools - lathe, shaping, milling, drilling, boring, and grinding machine, cutting tools, selection of cutting speed, feed and depth of cut for different operations. Machine drives and Work holding devices.

Production or manufacturing can be simply defined as value addition processes by which raw materials of low utility and value due to its inadequate material properties and poor or irregular size, shape and finish are converted into high utility and valued products with definite dimensions, forms and finish imparting some functional ability.

A machine is a physical system using power to apply forces and control movement to perform an action. The term is commonly applied to artificial devices, such as those employing engines or motors, but also to natural biological macromolecules, such as molecular machines

Basic functional principles of machine tool operations:

Machine Tools produce desired geometrical surfaces on solid bodies (preformed blanks) and for that they are basically comprised of;

- Devices for firmly holding the tool and work
- Drives for providing power and motions to the tool and work
- Kinematic system to transmit motion and power from the sources to the tool-work
- Automation and control systems
- Structural body to support and accommodate those systems with sufficient strength and rigidity.

Definition of Machining:

Machining is an essential process of finishing by which jobs are produced to the desired dimensions and surface finish by gradually removing the excess material from the preformed blank in the form of chips with the help of cutting tool(s) moved past the work surface(s).

Machining is process in which a cutting tool is used to remove small chips of material from the workpiece (the workpiece is often called the "work"). To perform the operation, **relative motion is required between the tool and the work.**

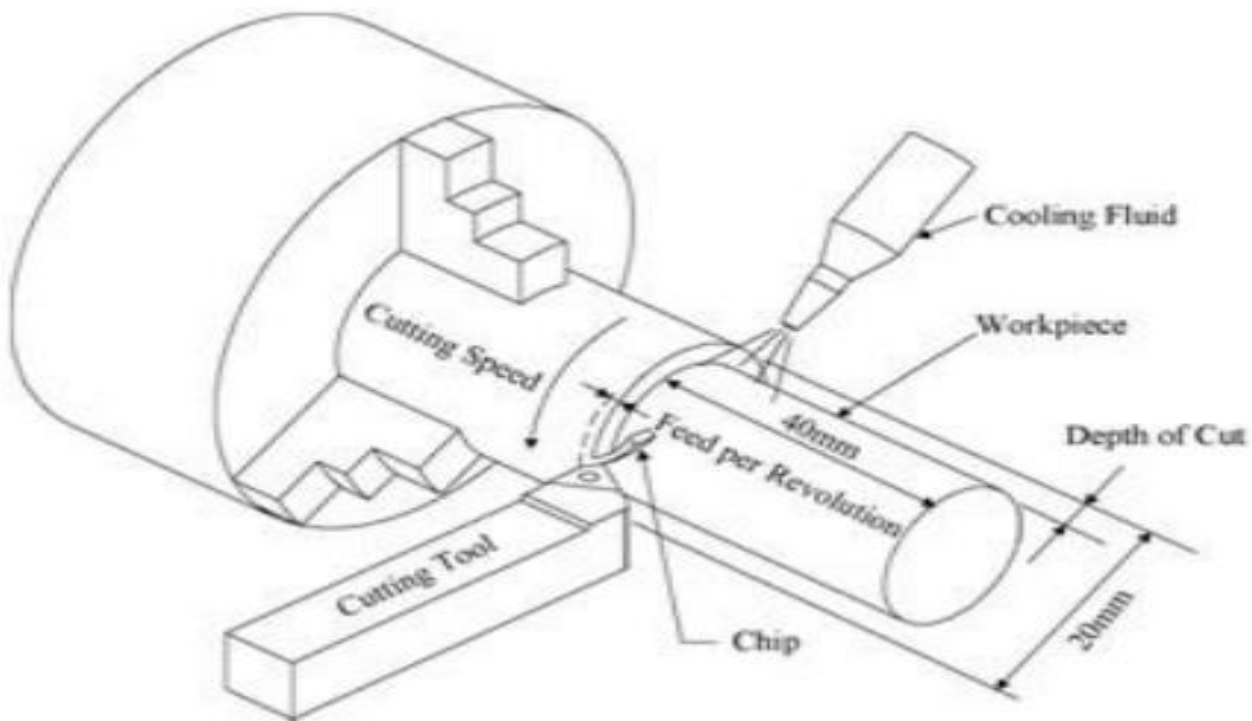
For material removal by machining, the work and the tool need relative movements and those motions and required power are derived from the power source(s) and transmitted through the kinematic system(s) comprised of a number and type of mechanisms.

In the turning process, the speed of chuck/spindle is set at certain **RPM (Rotations per minute)** to cut /machine surfaces of workpiece or blank.

In the turning process, the cutting tool is set at a certain **depth of cut (mm)** and travels to the left with a certain velocity as the workpiece rotates.

The **feed (mm/rev)**, is the distance the tool travels horizontally per unit revolution of the workpiece and **feed rate is in mm/min or m/sec**.

This movement of the tool produces a **chip**, which moves up the face of the tool.



Purpose of Machining:

Most of the engineering components such as gears, bearings, clutches, tools, screws and nuts etc. need dimensional and form accuracy and good surface finish for serving their purposes. Preforming like casting, forging etc. generally cannot provide the desired accuracy and finish. For that such preformed parts, called blanks, need semi-finishing and finishing and it is done by machining and grinding. Grinding is also basically a machining process.

Machining to high accuracy and finish essentially enables a product

- fulfill its functional requirements
- improve its performance
- prolong its service.

Basic functions of Machine Tools

Machine Tools basically produce geometrical surfaces like flat, cylindrical or any contour on the preformed blanks by machining work with the help of cutting tools. The physical functions of a Machine Tool in machining are:

- firmly holding the blank and the tool
- transmit motions to the tool and the blank
- provide power to the tool-work pair for the machining action.
- control of the machining parameters, i.e., speed, feed and depth of cut.

Machining operations

There are many kinds of machining operations, each of which is capable of generating a certain part geometry and surface texture.



In **turning**, a cutting tool with a single cutting edge is used to remove material from a rotating workpiece to generate a cylindrical shape. The primary motion is provided by rotating the workpiece, and the feed motion is achieved by moving the cutting tool slowly in a direction parallel to the axis of rotation of the workpiece.

Drilling is used to create a round hole. It is accomplished by a rotating tool that typically has two or four helical cutting edges. The tool is fed in a direction parallel to its axis of rotation into the workpiece to form the round hole.

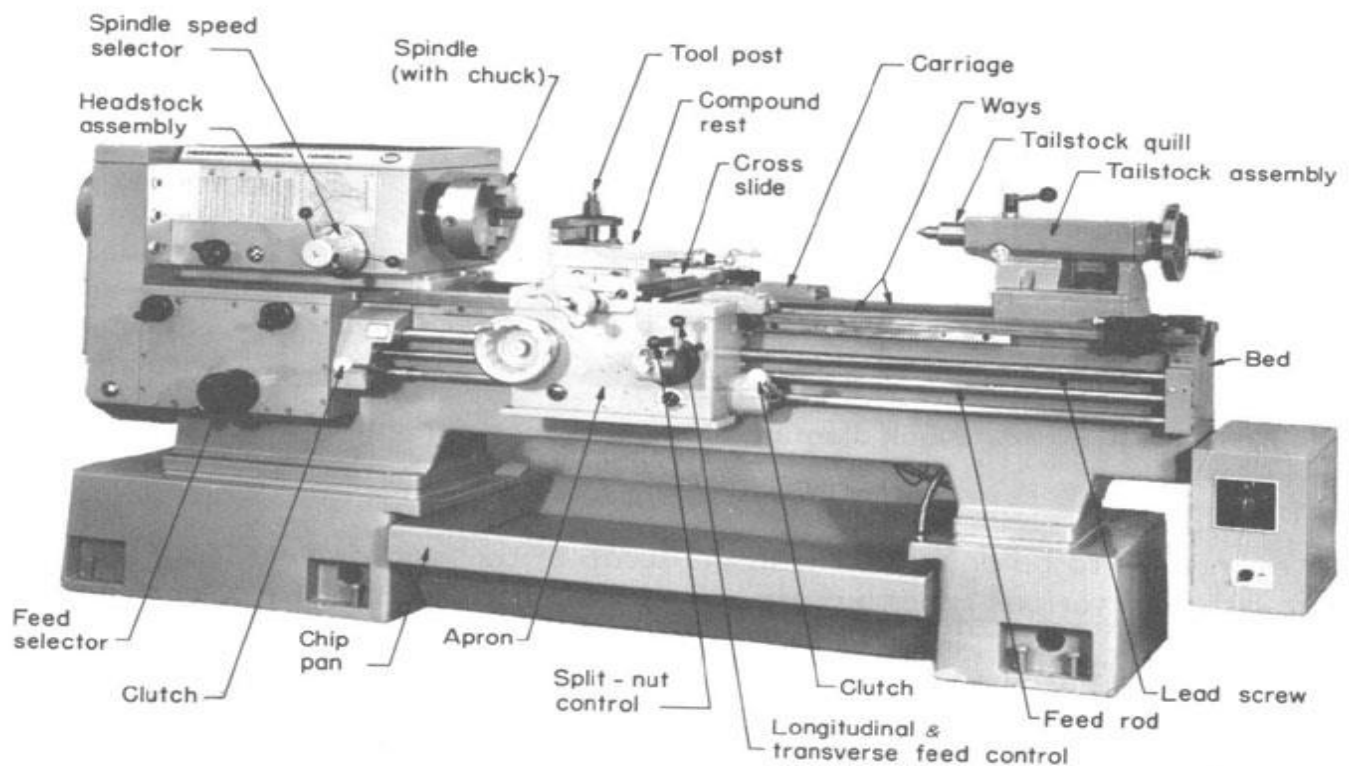
Shaping process in which cutting motion provided by linear-forward motion of the reciprocating tool and the intermittent feed motion provided by the slow transverse

motion of the workpiece resulting in a flat surface by gradual removal of excess material layer by layer.

In **boring**, a tool with a single bent pointed tip is advanced into a roughly made hole in a spinning workpiece to slightly enlarge the hole and improve its accuracy. It is a fine finishing operation used in the final stages of product manufacture.

In **milling**, a rotating tool with multiple cutting edges is moved slowly relative to the material to generate a plane or straight surface. The direction of the feed motion is perpendicular to the tool's axis of rotation. The speed motion is provided by the rotating milling cutter.

Lathe or Lathe machine



Split nut:

Split nut with the lead screw will be used for converting rotational movement into linear movement.

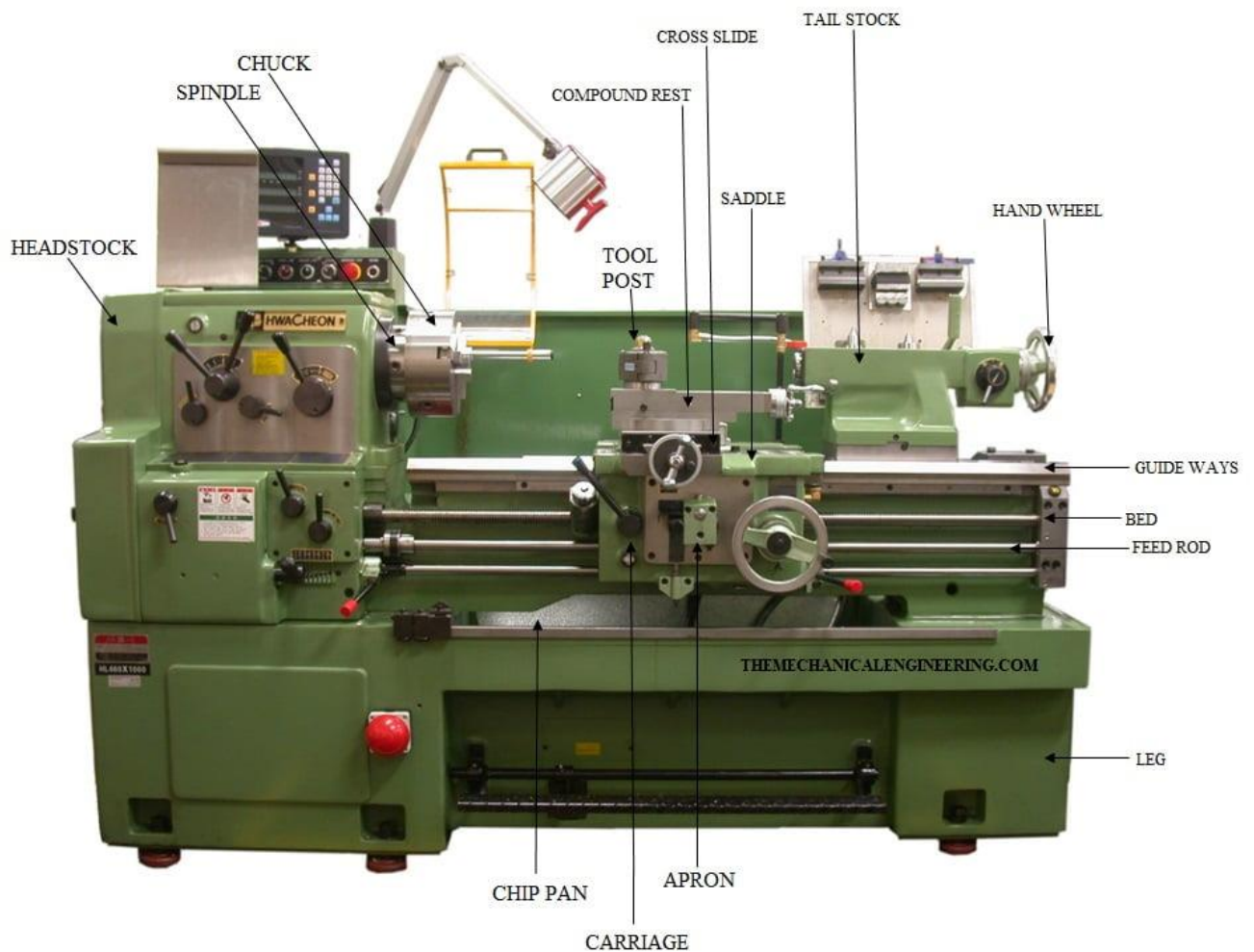
Chuck is classified into two types:

- **3-Jaw chuck:** It is used for cylindrical and circular shape jobs
- **4-Jaw chuck:** It is used for non-circular jobs.

Lathe Machine Operations:

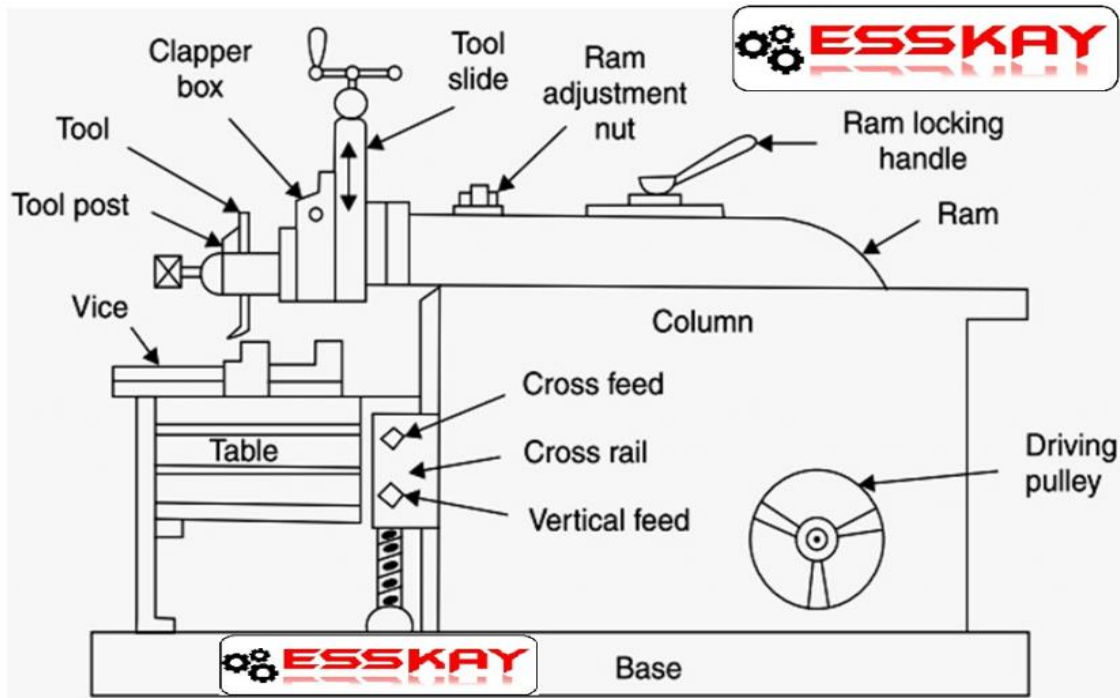
The operations which are performed on the lathe machine are called lathe operations and are as follows.

- Turning
- Facing
- Thread Cutting
- Drilling
- Boring
- Counterboring
- Reaming
- Many other operations

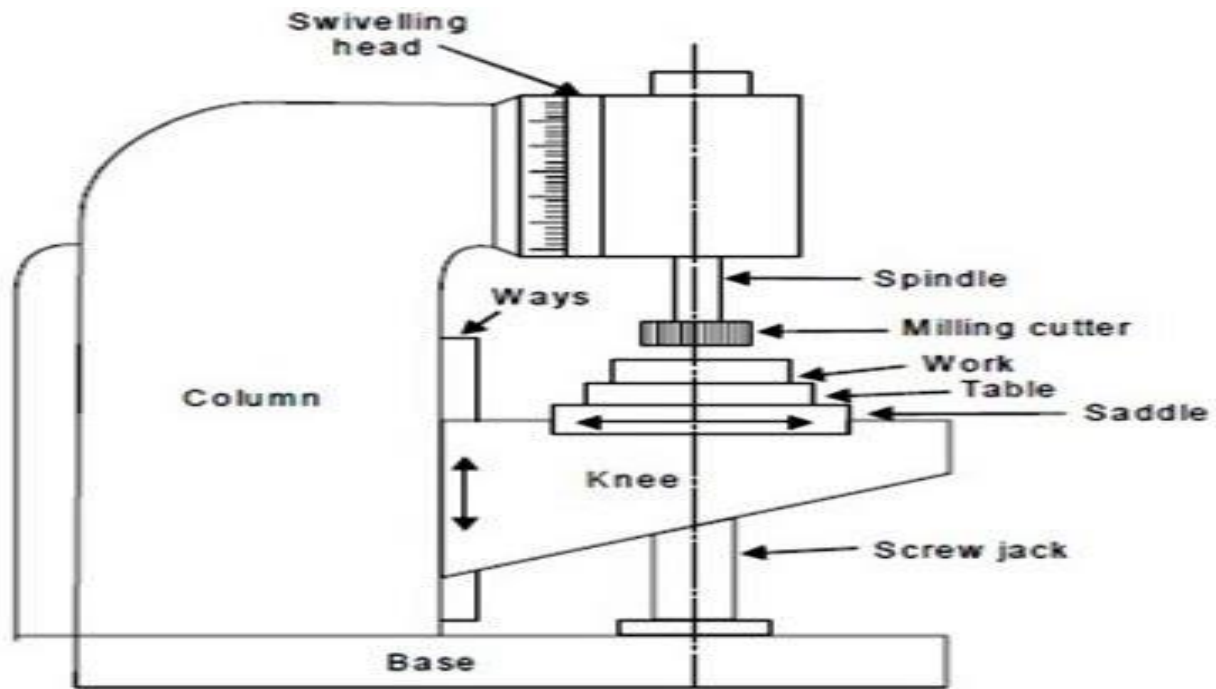


Shaping is a process in which cutting motion provided by linear-forward motion of the **reciprocating tool** and the intermittent feed motion provided by the slow transverse motion of the workpiece resulting in a flat surface by gradual removal of excess material layer by layer.

Shaping machine is a **machine designed for giving desired shapes to the surfaces** that may be horizontal, vertical and flat. A shaping tool is used to cut in curves, different angles, and many other shapes. A disc is responsible for the tool rotation which results in the forward and backward movement



Milling Machine



· A column and knee type milling machine

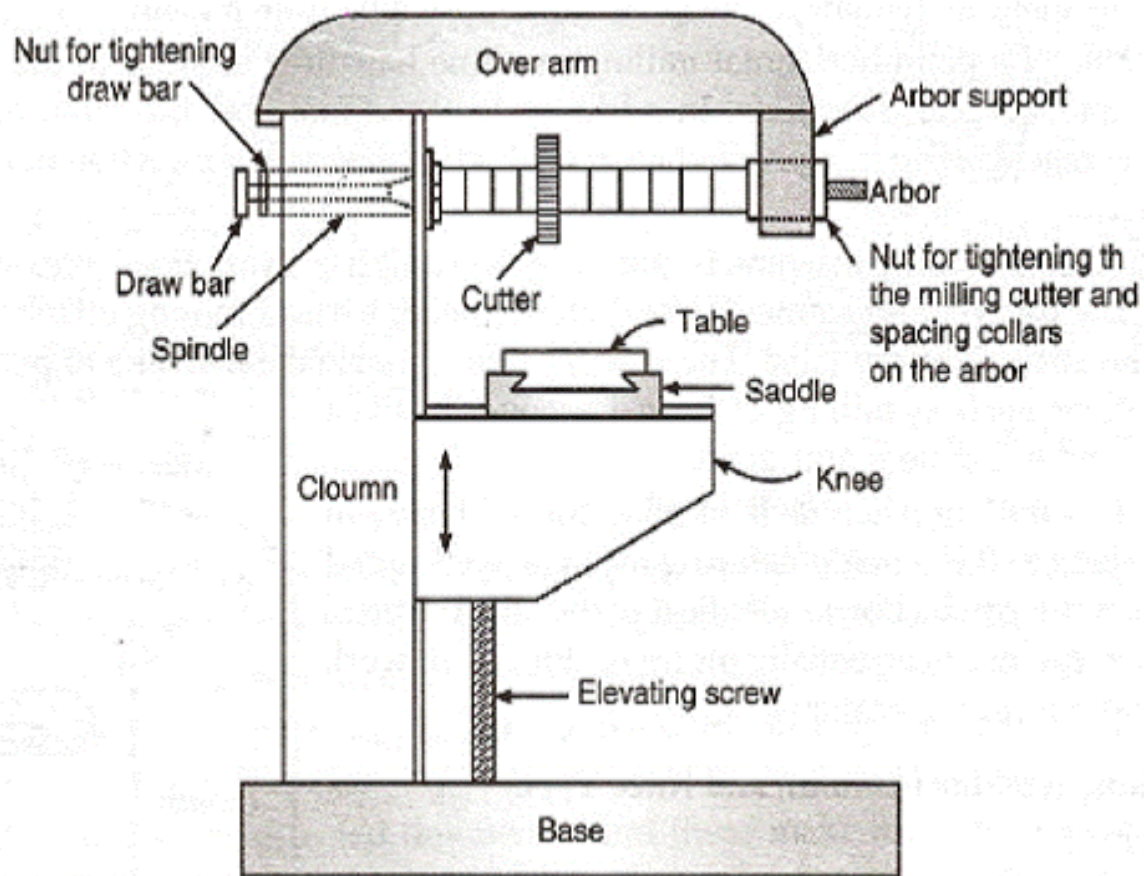
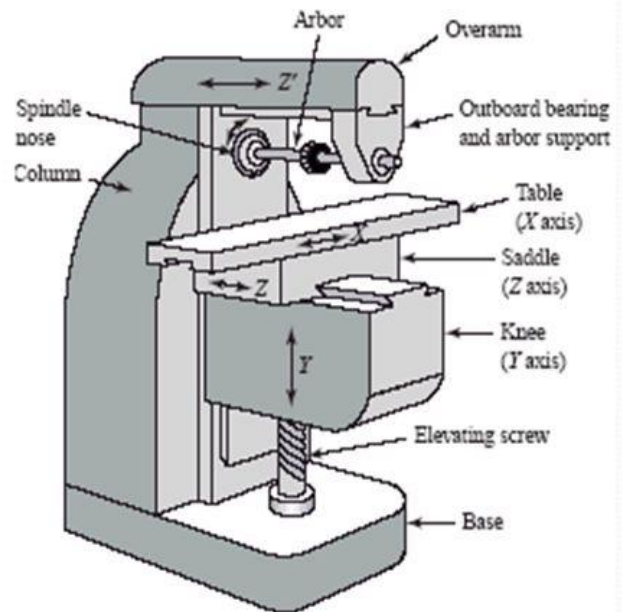
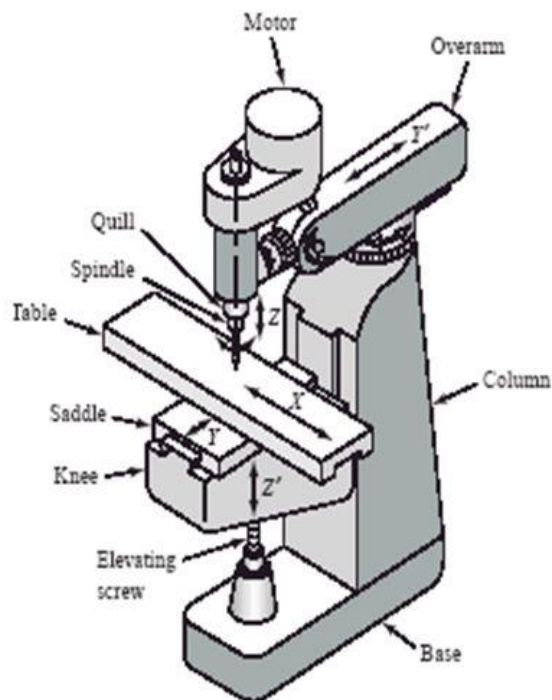
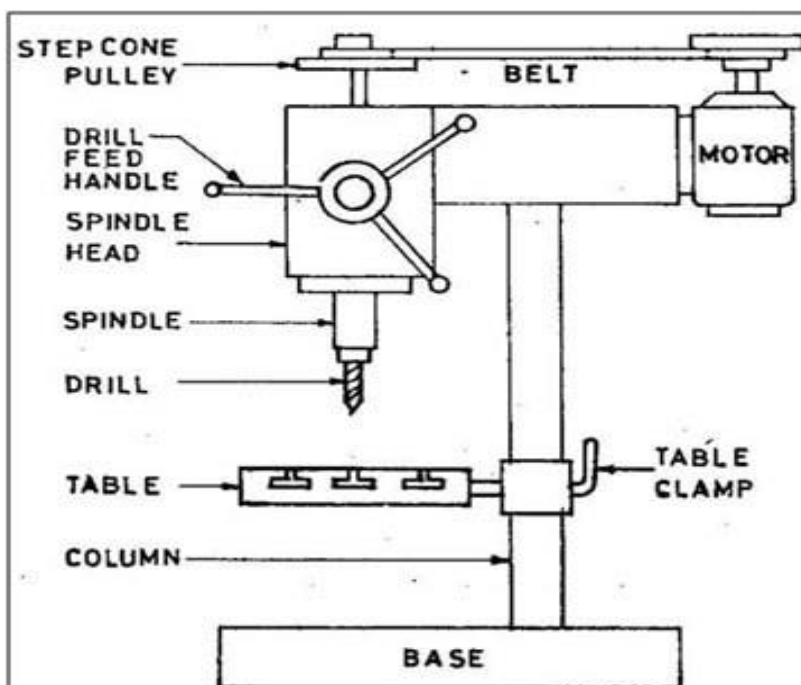


Diagram of milling machine



Drilling Machine

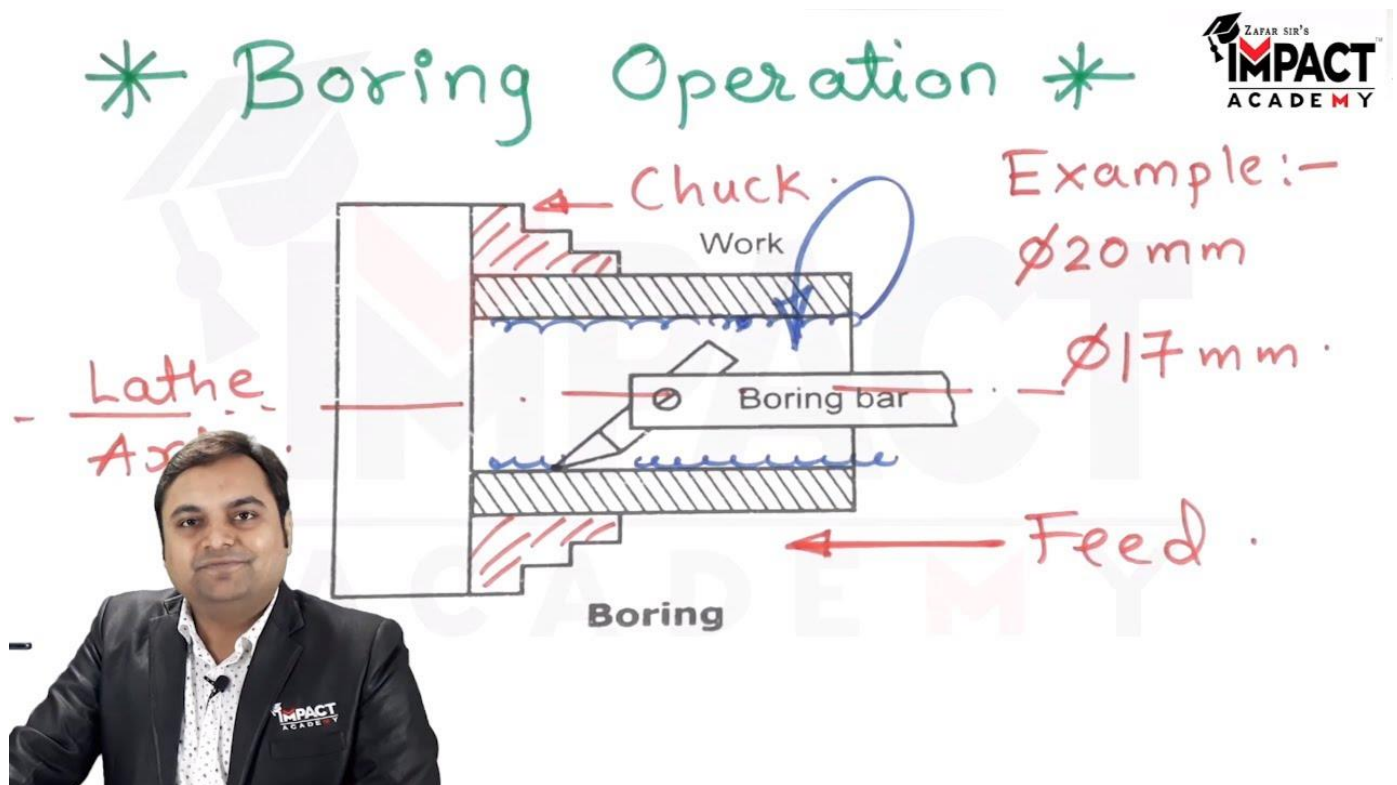


Boring Machine

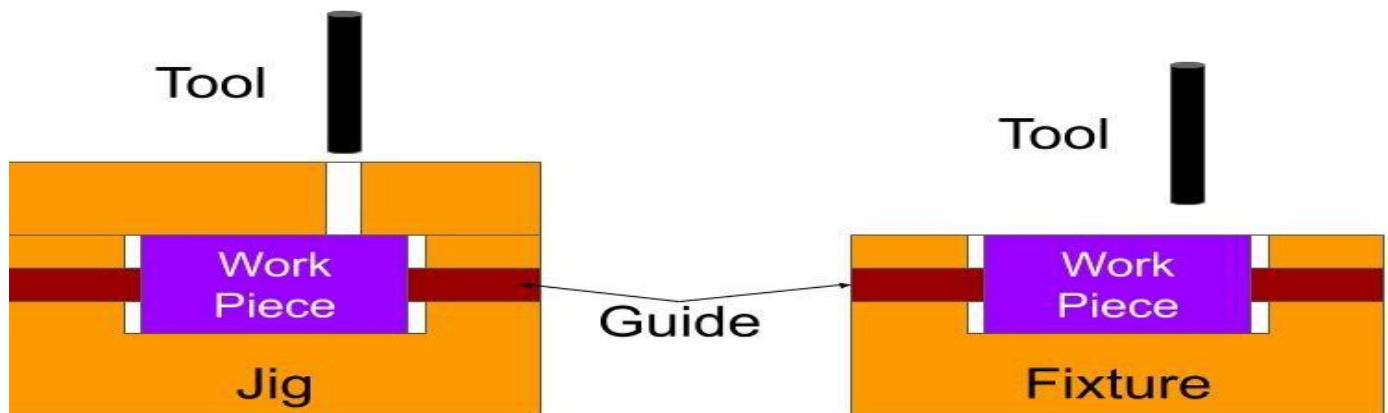
In machining, boring is the **process of enlarging a hole that has already been drilled (or cast)** by means of a single-point cutting tool (or of a boring head containing several such tools), such as in boring a gun barrel or an engine cylinder.

Boring is a cutting process that involves the use of a single-point cutting tool or **boring head to enlarge an existing hole in a workpiece**. This is in stark contrast to drilling, which is performed to create an initial hole in a workpiece.

Boring is a unit **process** in manufacturing as a mass reduction step, used for enlarging and accurately sized existing hole by means of single point cutting .



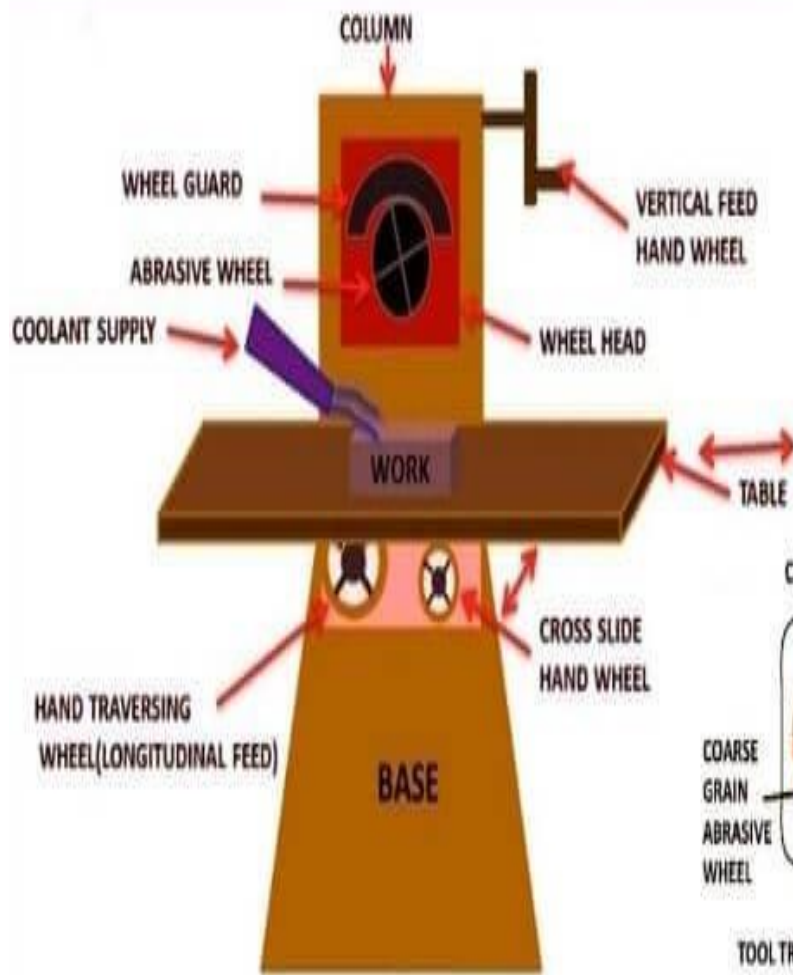
Jig Vs Fixture



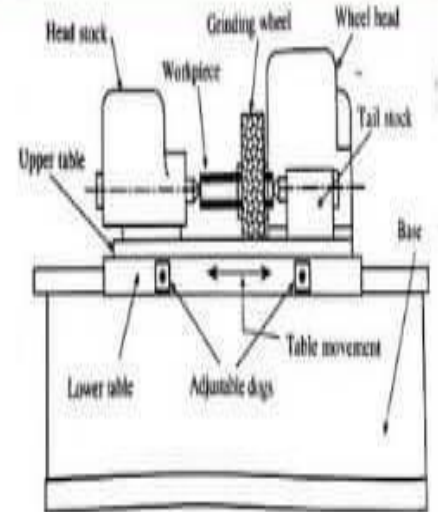
Grinding Machine

Grinding is the process of removing metal by the application of abrasives which are bonded to form a rotating wheel. When the moving abrasive particles contact the workpiece, they act as tiny cutting tools, each particle cutting a tiny chip from the workpiece.

Parts and Functions of Grinding Machine | Grinding Machine | Grinding Machine Types



Parts and Functions of Grinding Machine

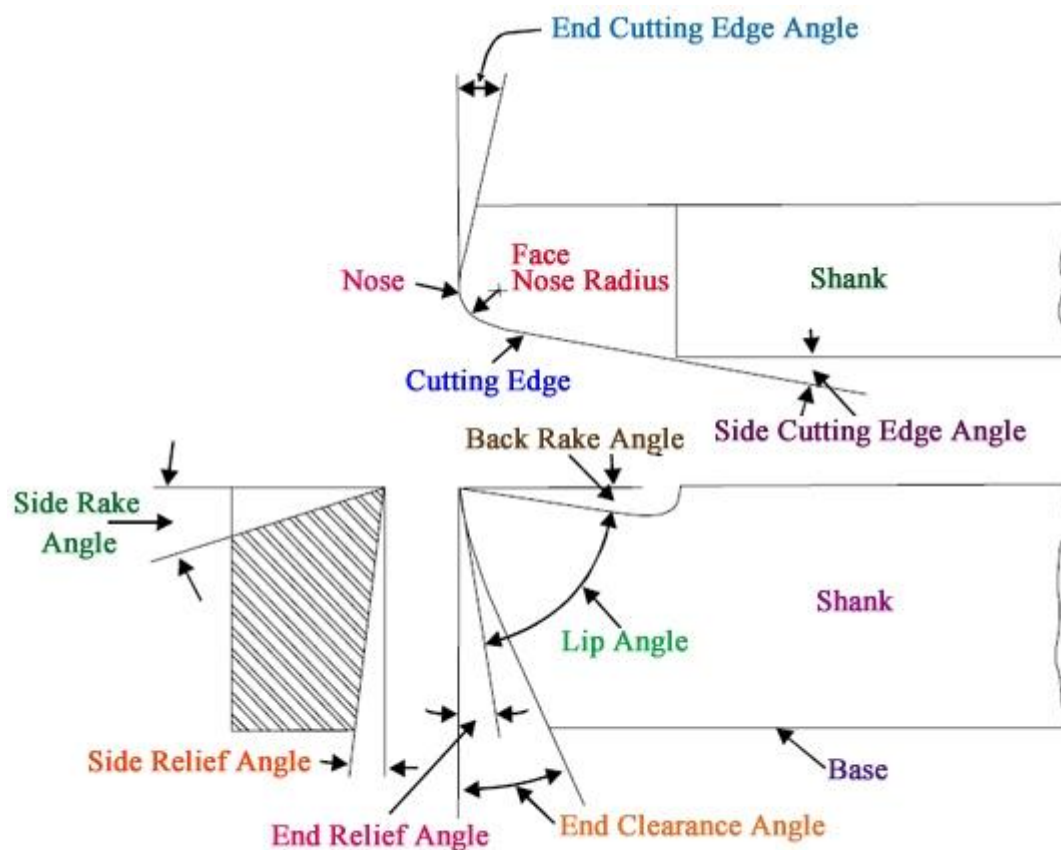
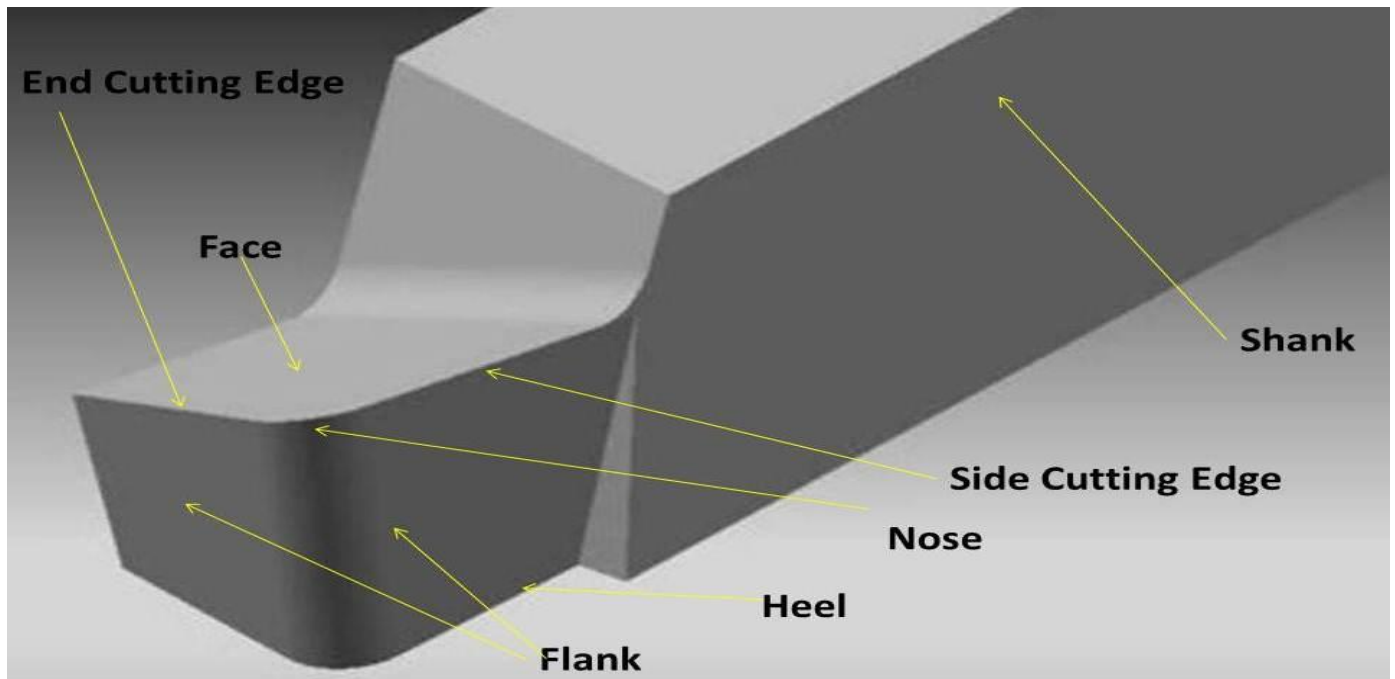


Cylindrical Grindings Machine

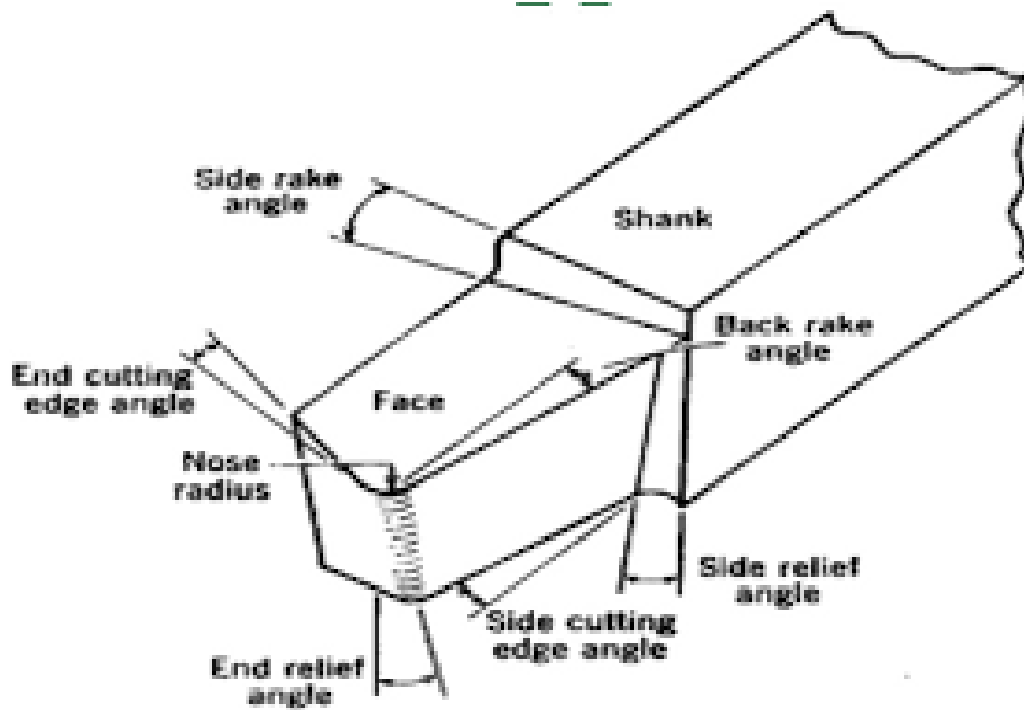


Grinding Machine

Cutting Tools



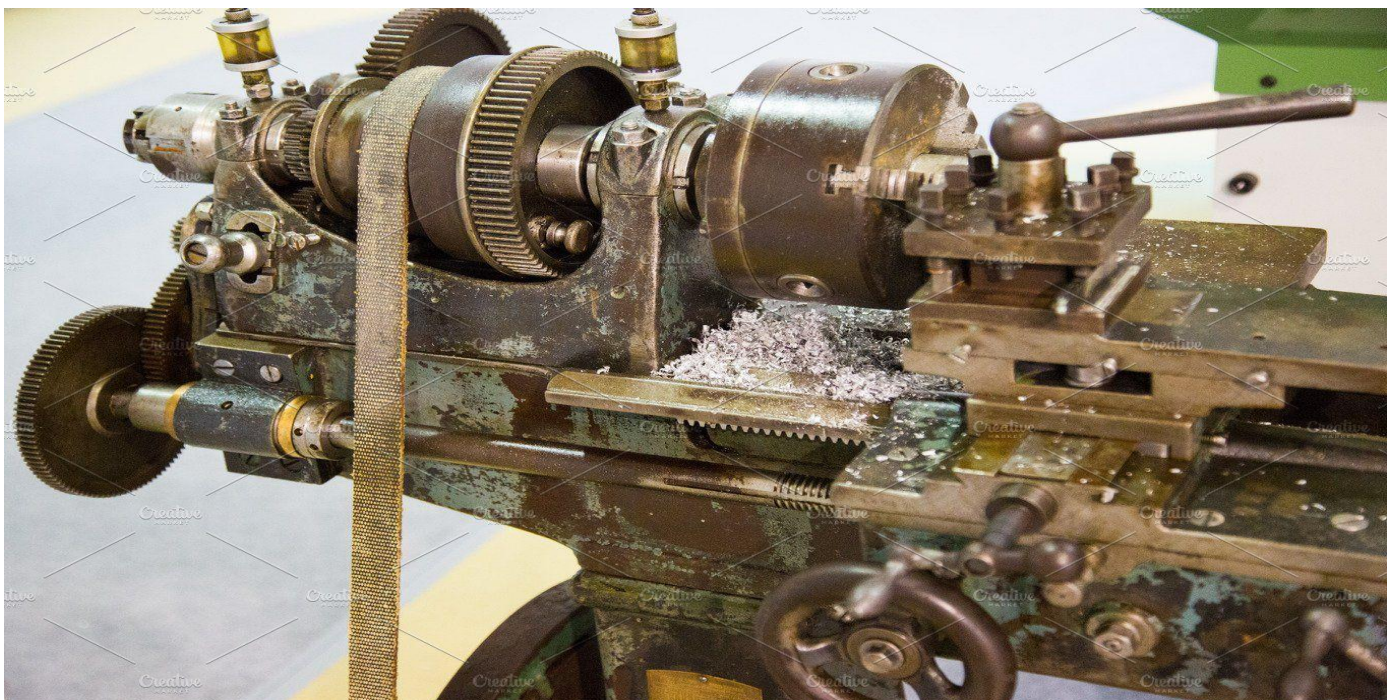
NOMENCLATURE OF SINGLE POINT CUTTING TOOL



Tool Nomenclature or Tool Signature or Tool Geometry of single point high speed cutting tool

1. Side Cutting edge angle 2. End cutting edge angle 3. Side relief angle 4. End relief angle 5. Back Rack angle 6. Side rack angle 7. Nose radius

Machine Tool Drives



Types of Drives - 1. Rotary Drives 2. Mechanical Drives 3. Belt Drives. 4. Chain Drive 6. Sprocket 7. Gearings

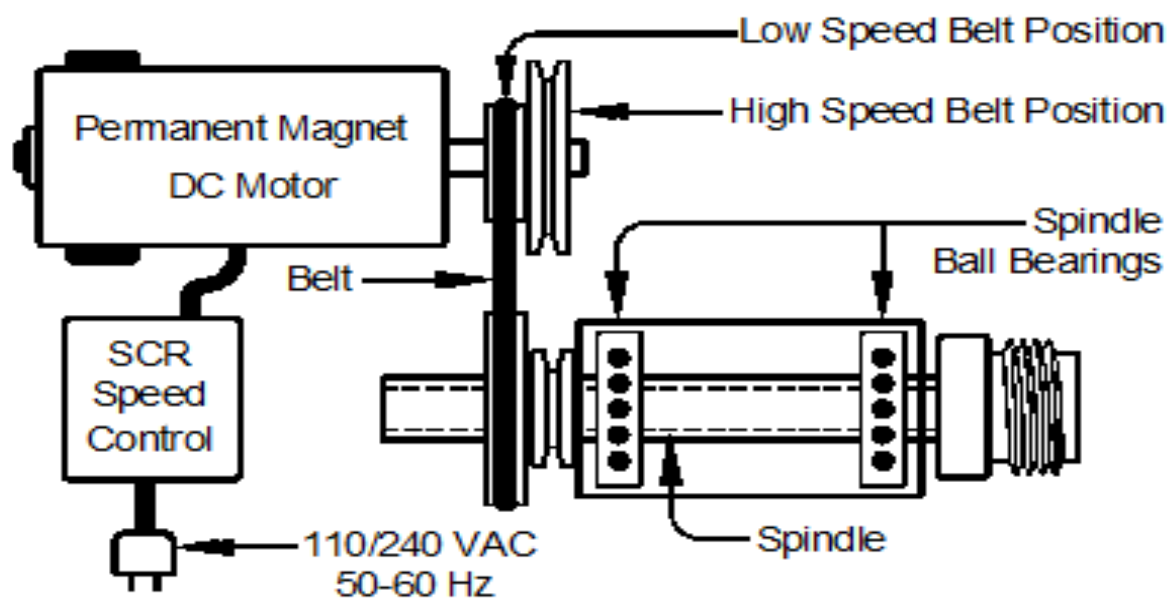
Drives are used to provide such kinds of controlled motion to the elements of a CNC machine tool. A drive system consists of drive motors and ball lead-screws. The control unit sends the amplified control signals to actuate drive motors which in turn rotate the ball lead-screws to position the machine table or cause rotation of the spindle. 2. Power drives Drives used in an automated system or in CNC system are of different types such as electrical, hydraulic or pneumatic.

- **Electrical drives:** These are direct current (DC) or alternating current (AC) servo motors. They are small in size and are easy to control.
- **Hydraulic drives:** These drives have large power to size ratio and provide stepless motion with great accuracy. But these are difficult to maintain and are bulky. Generally they employ petroleum based hydraulic oil which may have fire hazards at upper level of working temperatures. Also hydraulic elements need special treatment to protect them against corrosion.
- **Pneumatic drives:** This drives use air as working medium which is available in abundant and is fire proof. They are simple in construction and are cheaper. However these drives generate low power, have less positioning accuracy and are noisy.

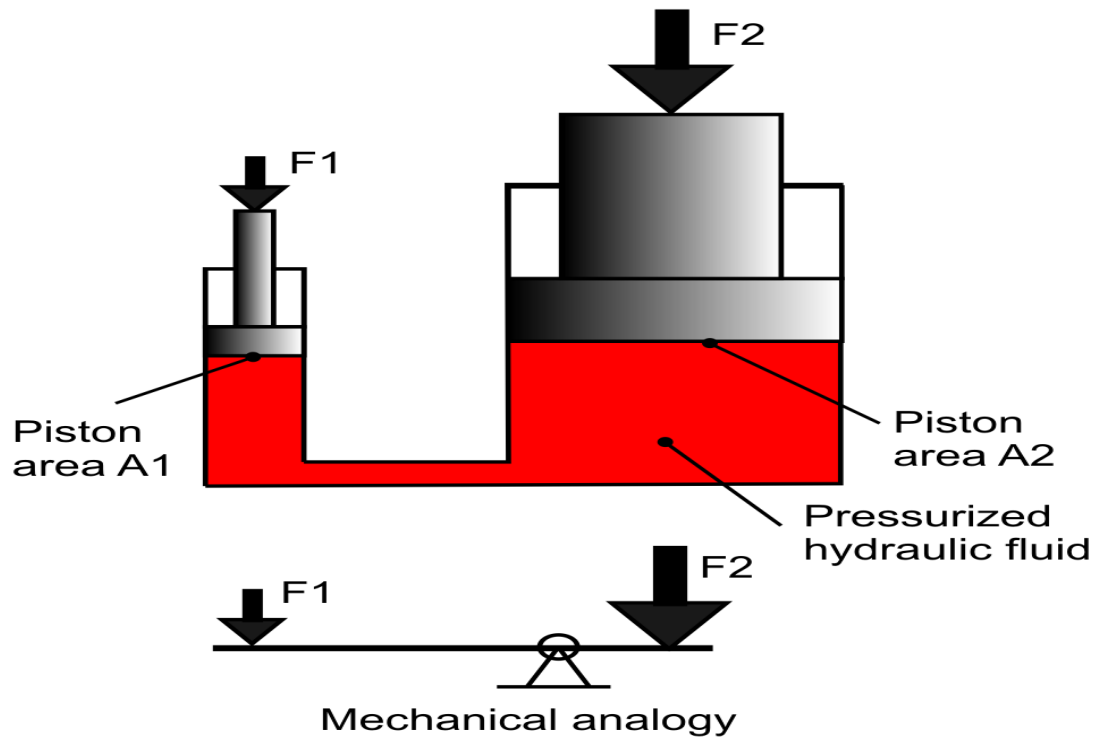
In CNC, usually AC, DC, servo and stepper electrical drives are used. The various drives used in CNC machines can be classified as: a. Spindle drives to provide the main spindle power for cutting action b. Feed drives to drive the axis.

(i) **Stepper motor** is a pulse-driven motor that changes the angular position of the rotor in steps. Due to this nature of a stepper motor, it is widely used in low cost, open loop position control systems.

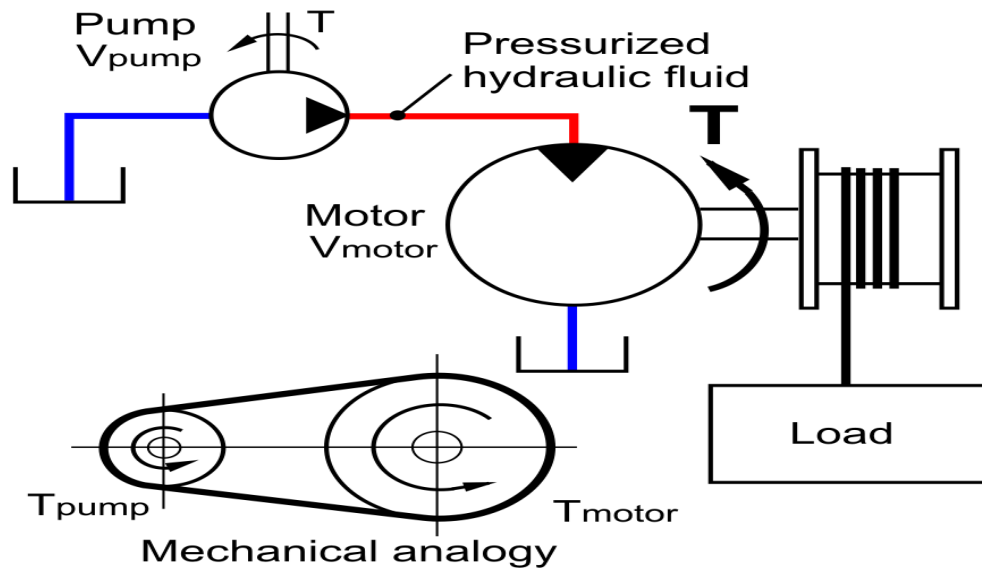
(ii) **Servomotor:** Servomotors are special electromechanical devices that produce precise degrees of rotation. A servo motor is a DC or AC or brushless DC motor combined with a position sensing device. Servomotors are also called control motors as they are involved in controlling a mechanical system. The servomotors are used in a closed-loop servo system as shown in Figure 4.2.4. A reference input is sent to the servo amplifier, which controls the speed of the servomotor. A feedback device is mounted on the machine, which is either an encoder or resolver. This device changes mechanical motion into electrical signals and is used as a feedback. This feedback is sent to the error detector, which compares the actual operation with that of the reference input. If there is an error, that error is fed directly to the amplifier, which will be used to make necessary corrections in control action. In many servo systems, both velocity and position are monitored. Servomotors provide accurate speed, torque, and have ability of direction control.



Force increase with hydraulics
 $F_2 = F_1 \cdot (A_2/A_1)$



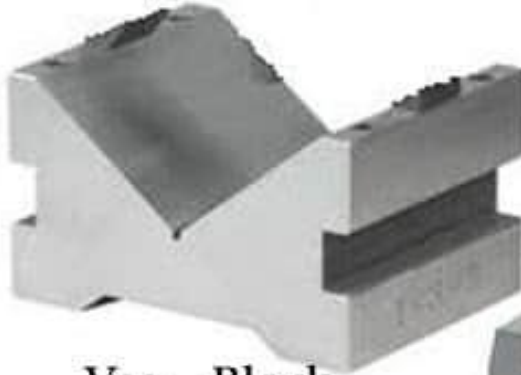
Torque increase with hydraulics
 $T_{\text{motor}} = (V_{\text{motor}}/V_{\text{pump}}) \cdot T_{\text{pump}}$



Work holding devices



Angle Plate



Vee - Block



Swivel Vice



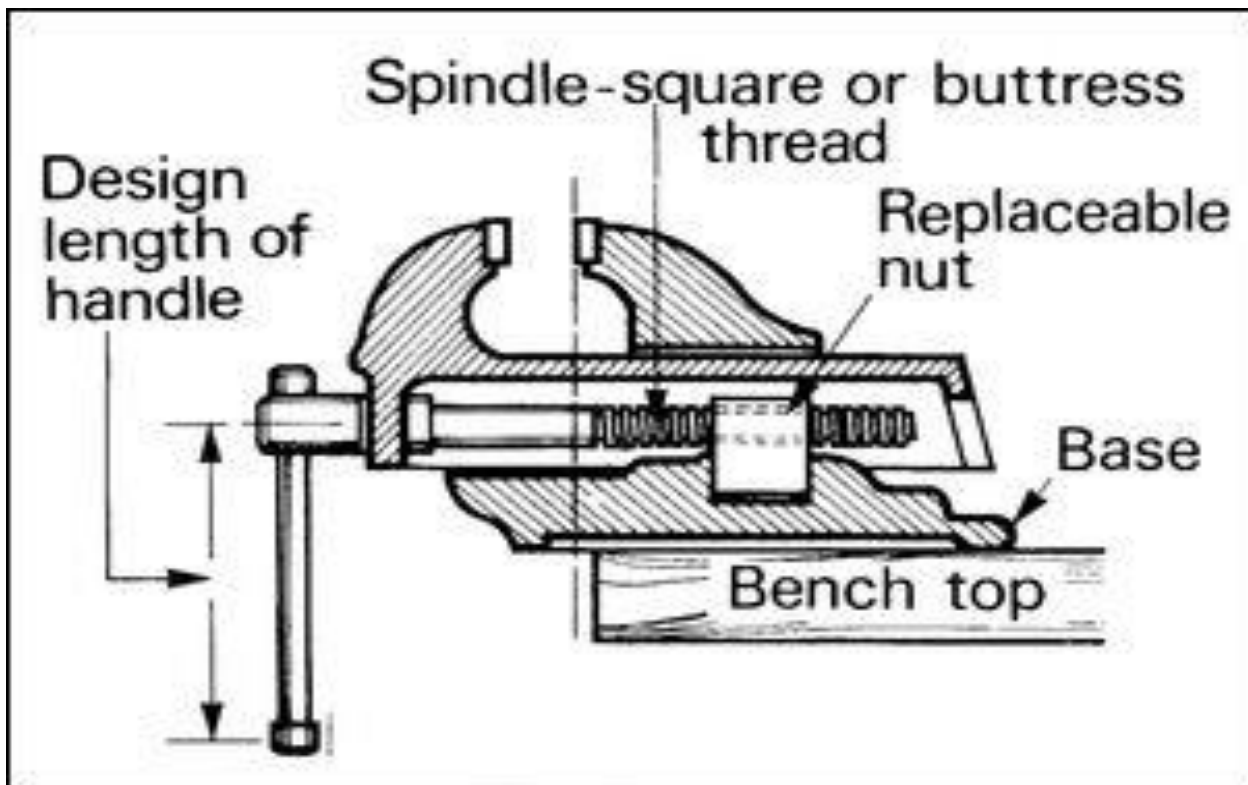
Plain Vice

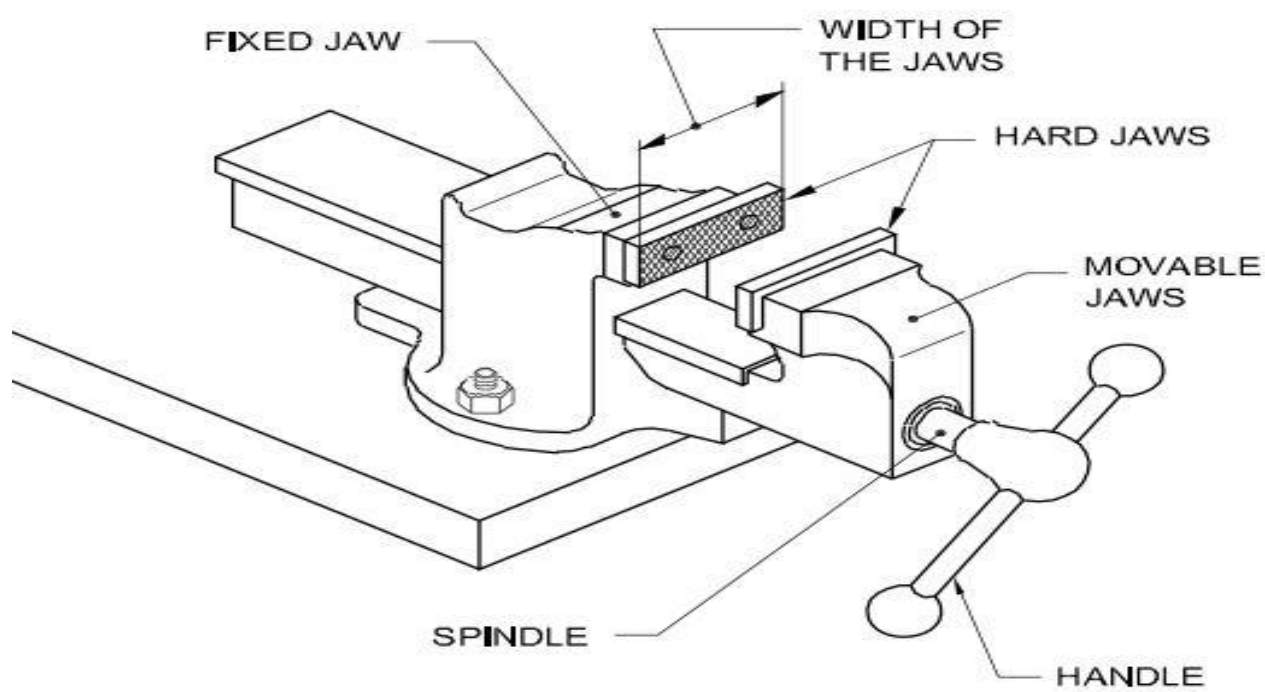
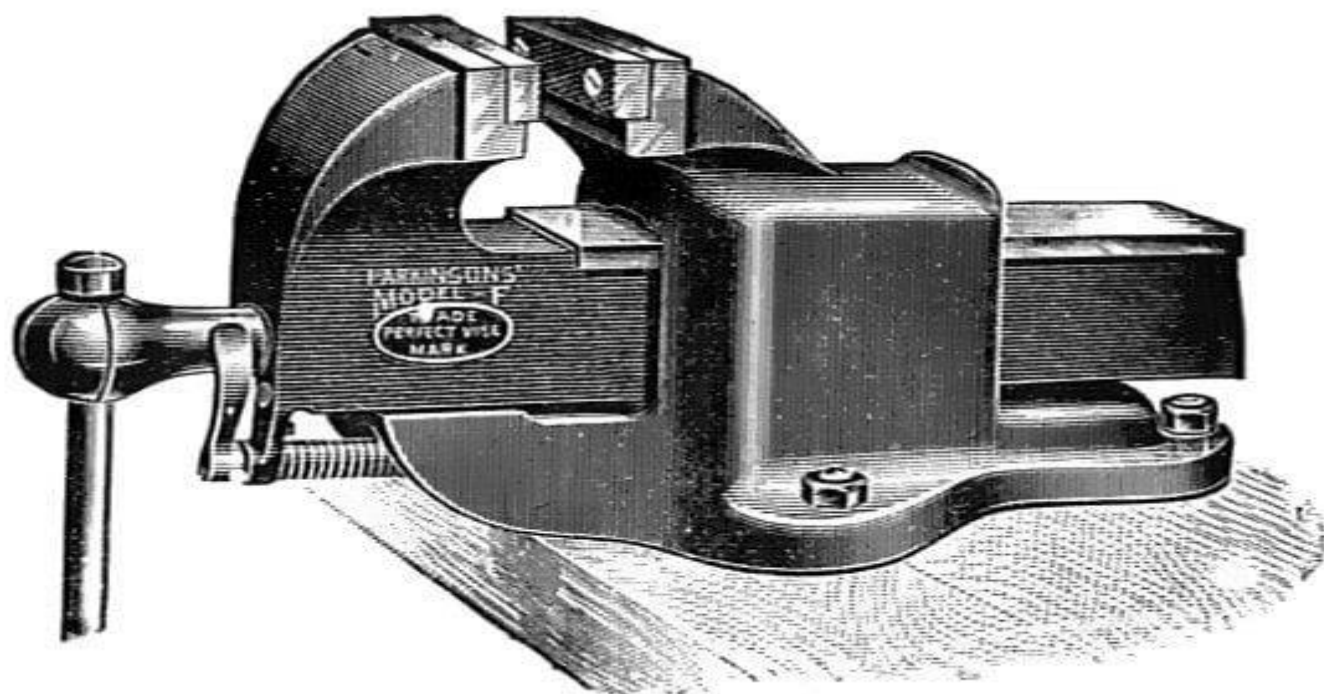


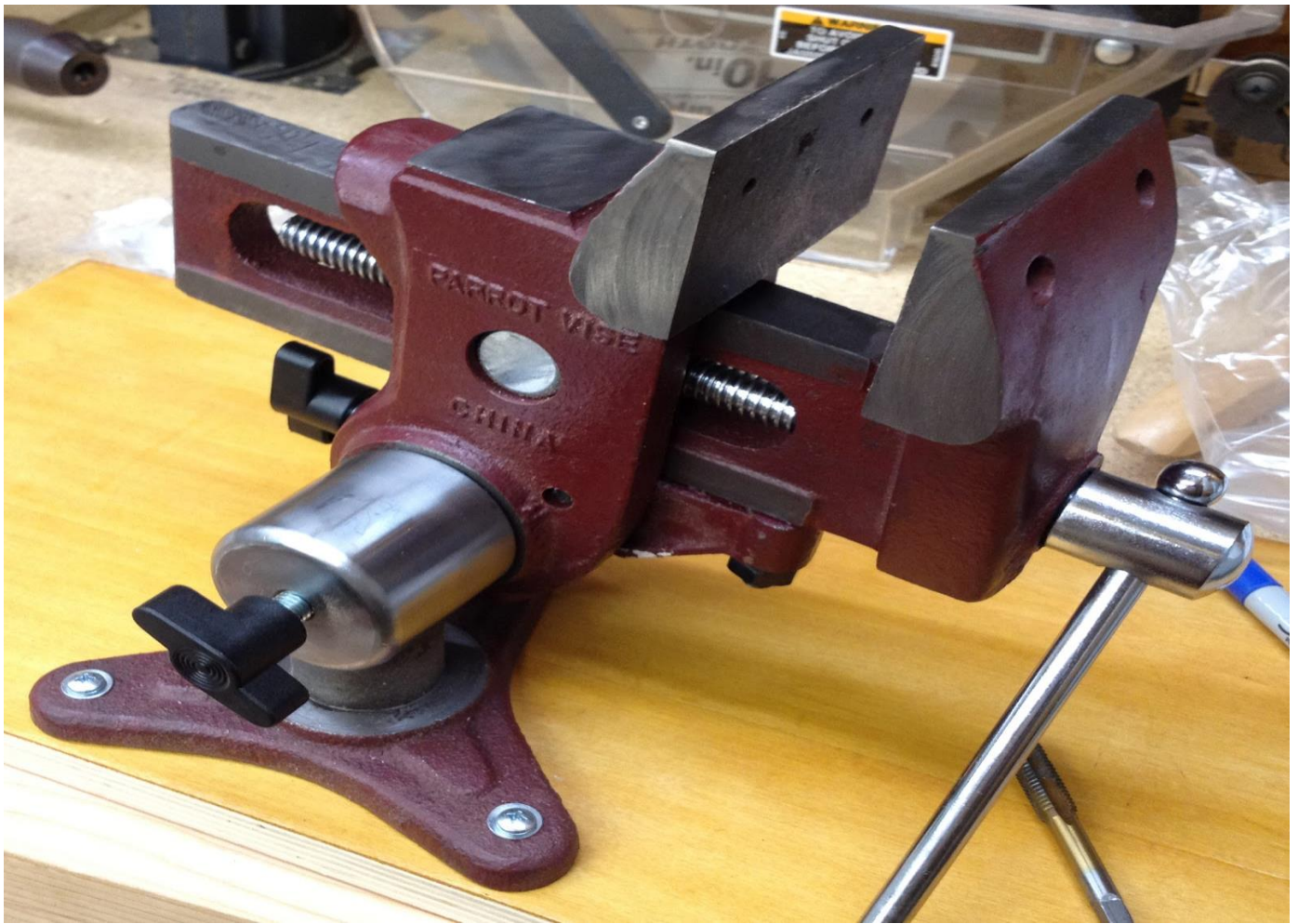
Universal Vice

Work Holding Devices Used on Milling Machine

Fitting vice

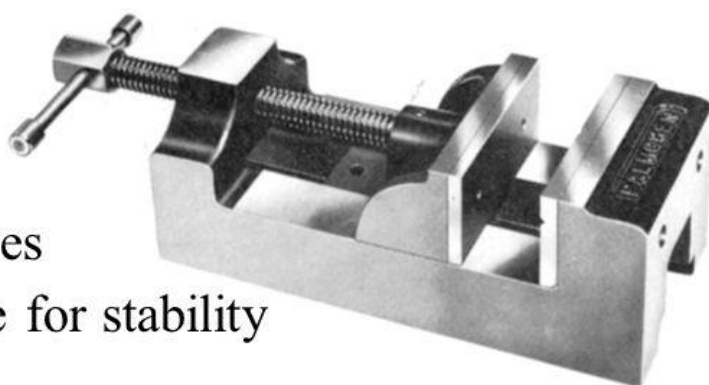




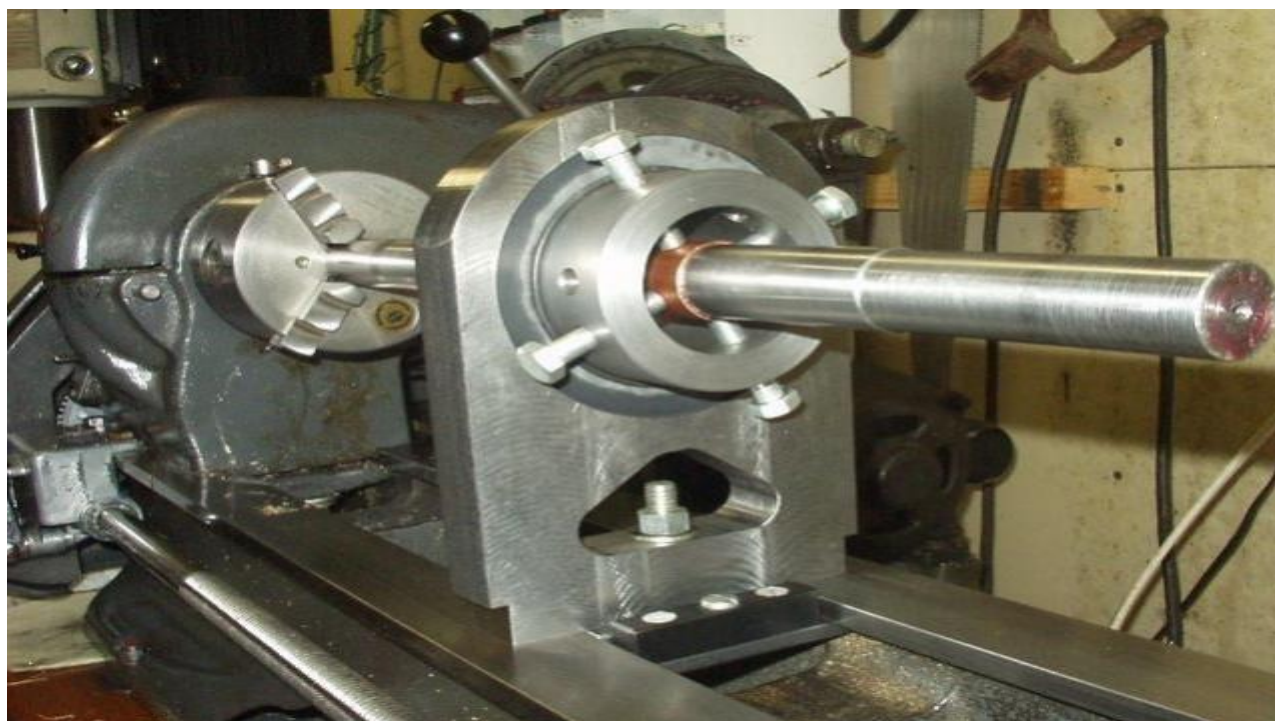


Work-Holding Devices

- Angle vise
 - Angular adjustment on base to allow operator to drill holes at an angle without tilting table
- Drill vise
 - Used to hold round, square or odd-shaped rectangular, pieces
 - Bolt vise to table for stability



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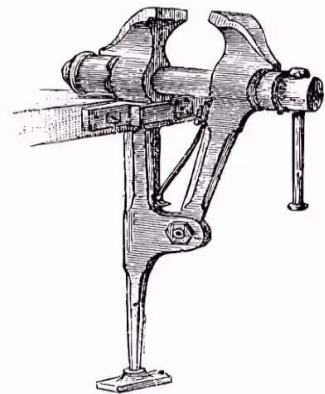
Bench Vice



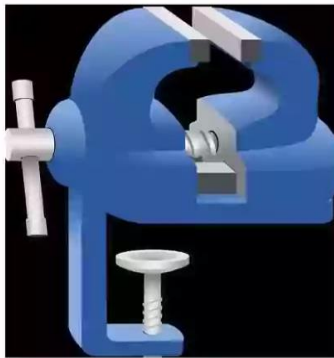
Pipe Vice



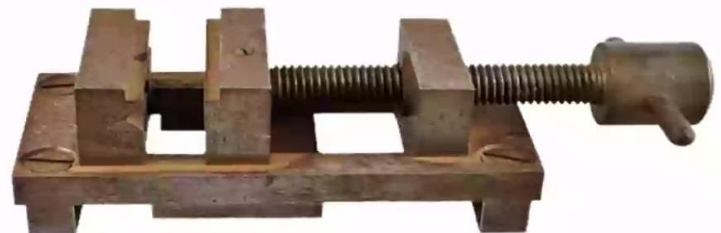
Tool Makers Vice



Hand Vice



Leg Vice



Drill Vice



Selection of cutting speed, feed and depth of cut for different operations

Cutting Speed :

Cutting Speed is the speed at which the metal is removed by the cutting tool from the workpiece. In the case of lathe machine cutting speed is the peripheral speed of the work past the cutting tool. It is expressed in meter/min. or mm/min.

Cutting speed depends upon the following factors:

- i. Tool material.
- ii. Work material.
- iii. Depth of cut.
- iv. Tool geometry.
- v. Type of machine tool.
- vi. Surface quality required.

Feed rate :

Feed rate is defined as the distance the tool travels during one revolution of the part. Cutting speed and feed determines the surface finish, power requirements, and material removal rate. The primary factor in choosing feed and speed is the material to be cut. However, one should also consider the material of the tool, rigidity of the workpiece, size and condition of the lathe, and depth of cut. For most Aluminum alloys, on a roughing cut (.010 to .020 inches depth of cut) run at 600 fpm. On a finishing cut (.002 to .010 **depth of cut**) run at 1000 fpm. To calculate the proper spindle speed, divide the desired cutting speed by the circumference of the work. Experiment with feed rates to achieve the desired finish. In considering the depth of cut, it's important to remember that for each thousandth depth of cut, the working diameter is reduced by two thousandths.

Depth of Cut

The depth of cut is the distance that the tool bit moves into the work. usually measured in thousandths of an inch or millimeters. General machine practice is to use a depth of cut up to five times the rate of feed, such as rough cutting stainless steel using a feed of 0.020 inches per revolution and a depth of cut of 0.100 inch. which would reduce the diameter by 0.200 inch. If chatter marks or machine noise develops, reduce the depth of cut.

It is the total amount of metal removed per pass of the cutting tool. It is expressed in mm. It can vary and depend upon the type of tool and work material.

Mathematically, it is half of the difference of diameters.

Suggested Cutting Speeds for Machining

<i>Material</i>	<i>Cutting Speed, $\frac{m}{min}$</i>				
	<i>Turning</i>		<i>Drilling</i>	<i>Milling</i>	
	<i>HSS</i>	<i>Carbide</i>	<i>HSS</i>	<i>HSS</i>	<i>Carbide</i>
Mild steel	20-35	90-135	20-30	20-35	75-130
Aluminum	150-180	335-365	30-120	150-180	335-365
Free machining brass	90-105	180-200	50-55	90-105	180-200
Other brass & bronze	60-75	150-165	35-40	60-75	150-165
Copper & phosphor bronze	30-35	60-65	20	30-35	60-65
Wood & plastic	60-150	150-450	20-90	120	180-450
Unhardened stainless steel	20-35	70-120	15-25	15-30	60-115
Class 20 cast iron	35	135	30	130	30

HSS: High speed steel

Carbide: Cemented carbide

Sources: *Machinery's Handbook*, Kalpakjian & Schmid textbook, M. Chapin

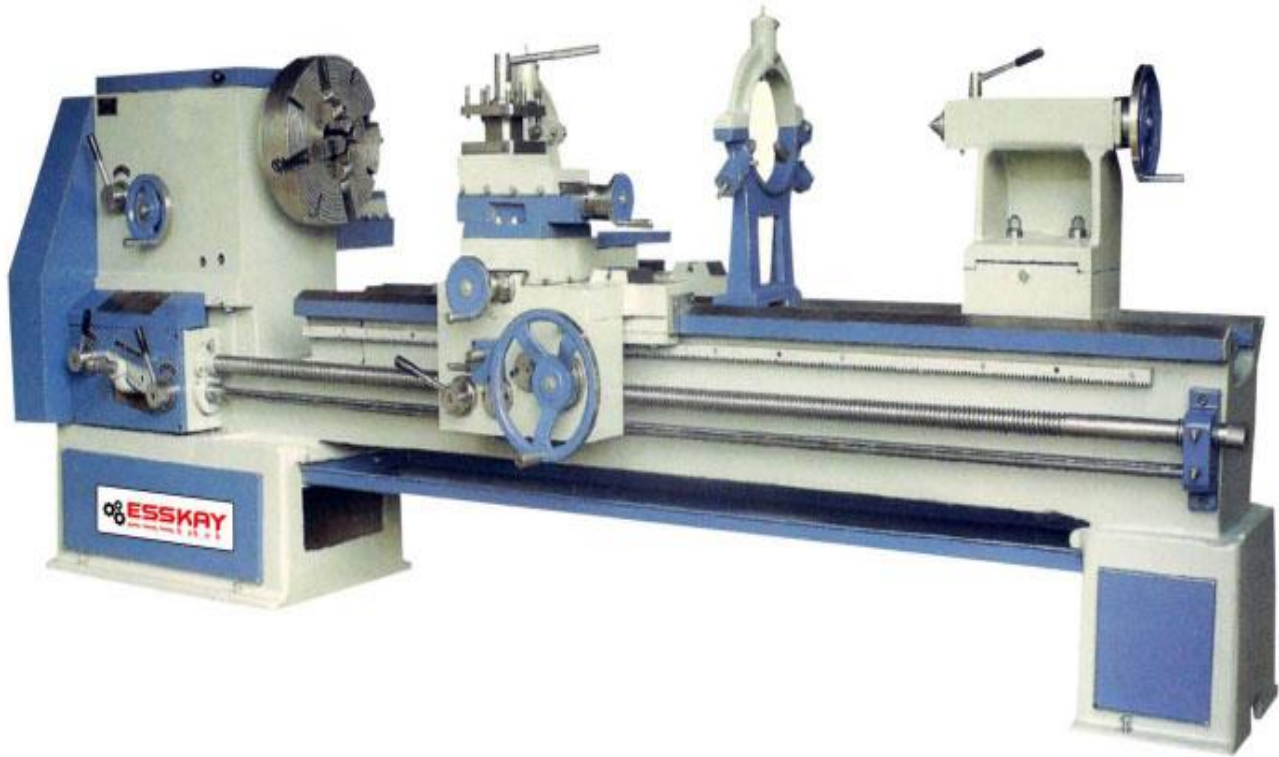
AVERAGE CUTTING SPEED, FEED AND DEPTH OF CUT FOR DIFFERENT TOOL MATERIALS

Material	H.S.S v.m.p.m.	Stellite v.m.p.m.	Cemented Carbide v.m.p.m.	Feed Mm/rev	Depth of Cut mm
Cast Iron	15 – 19	30	63	0.2 to 0.8	0.5 to 1 for finishing operations 2 to 5 for roughing operation
Mild Steel	25 – 31	55	810		
Brass	60 – 90	120	180		
Aluminium	120	300	360		

Stellite® 6, the nominal chemical composition of which is 27–32% Cr, 4–6% W, 0.9–1.4% C, with additions of Ni, Fe, Si, Mn and Mo and Co as balance, is a two-phase alloy, with a chromium carbide minor phase, which has been shown to have outstanding resistance to both erosion and corrosion.

Specification of Lathe machine

Specification is a document stating all requirements of a product, process, system or service.



SPECIFICATION OF LATHE MACHINE

MODEL	EKL - 614	EKL - 816	EKL - 1018	EKL - 1220	EKL - 1222	EKL- 1224
Length of bed	1828 MM	2438 MM	3048 MM	3657 MM	3657 MM	3657 MM
Width of bed	350 MM	400 MM	450 MM	500 MM	550 MM	600 MM
Height of Center	300 MM	350 MM	400 MM	450 MM	500 MM	550 MM
Swing Over Slide	430 MM	533 MM	635 MM	736 MM	787 MM	863 MM
Swing in Gap	914 MM	1016 MM	1117 MM	1370 MM	1422 MM	1473 MM
Admit Between Centers	838 MM	1371 MM	1980 MM	2413 MM	2413 MM	2363 MM
Spindle Hollow	53 MM	63 MM	88 MM	103 MM	103 MM	103 MM
Type of Bed	Pedestal	Pedestal	Pedestal	Planner	Planner	Planner
Hp. Required	2*960	2*960	3*960	3*960	3*960	5*960
T.P.L of Lead Screw	4	2	2	2	2	2

Technical Specification	Banka 34	Banka 35	Banka 40	Banka 43
Height of Centers	170	175	200	215
Swing over bed	340	350	400	430
Swing over slide	180	190	240	270
Width of Bed	240	242	280	305
Spindle bore	40	40	51	52
Spindle nose	LOO-Type	LOO-Type	LO-Type	LO-Type
Spindle Speed No.	8	16 H-L	16 H-L	16 H-L
RPM	60-1025	48-1535	48-1535	40-1415
Cross slide Travel	175	180	210	240
Top slide Travel	125	125	125	130
Tail Stock Sleeve	MT-3	MT-3	MT-3	MT-4
Quill Dia. /Travel	50/125	50/125	50/125	75/150
Tool shank section	20 X 20mm	20 X 20mm	20 X 20mm	20 X 20mm
Feeds Longitudinal No.	60/0.08-1	60/0.08-1	60/0.08-1	60/0.08-1
Feeds Transverse No.	60/0.02-0.25	60/0.02-0.25	60/0.02-0.25	60/0.02-0.25
Metric Thread No.	40/0.4-7mm	40/0.4-7mm	40/0.4-7mm	40/0.4-7mm
Thread Per Inch No.	45/70-4 TPI	45/70-4 TPI	45/70-4 TPI	45/70-4 TPI
Lead Screw	4 TPI	4 TPI	4 TPI	4 TPI
Main Motor	1.5kw	1.5kw	1.5kw	2.25kw
Admit Between Center	600 / 750	750/1000	750/1000	1000/1500/2500
Bed Length Feet	4.5' / 5'	5' / 6'	5' / 6'	6' 8' / 11'
Weight (Kg) Approximate	600 / 650	750/800	850/900	1000/1200/1500

According to the materials used the cutting tools are classified into

1. Carbon tool steel
2. High speed steel tool (HSS)
3. Cemented carbide
4. Ceramics tool
5. Cubic boron nitride Tool (CBN)
6. Diamond tool

Carbon tool steel

Carbon tool steel is one of the inexpensive metal cutting tools used for the low-speed machining operation. These plain carbon steel cutting tool have the composition of 0.6-1.5% carbon and very small amount of (less than 0.5 %) Mn, Si. Other metal like Cr, V are added to change the hardness and grain size. High carbon steels are abrasion resistant and have the ability to maintain sharp cutting edge. Carbon tool steels possess good [machinability](#). This material loses their hardness rapidly at a temperature about 250°C. Therefore, it can't use high-temperature application. It does not prefer in a modern machining operation.

Carbon steel tool is used in [twist drills](#), milling tools, turning and forming tools, used for soft

material such as brass, aluminum magnesium, etc.

Temperature - 450°C

Hardness – up to HRC 65

High-speed steel (HSS) (W: Cr: V -18 : 4: 1)

This is a high carbon steel with a significant amount of alloying element, such as tungsten, molybdenum, chromium, etc. to improve hardenability, toughness and wear resistance. It gives a higher metal removal rate. It loses its hardness at a moderate temperature about 650°C. Therefore, a coolant should be used to increase [tool life](#). It can use many times by re-sharpening. Some surface treatment is done on the HSS to improve its property.

Surface treatment used in the HSS

Super finishing - Reduce friction

Nitriding - Increase wear resistance

Chromium electroplating - Reduce friction

Oxidation - Reduce friction

High-speed steel tools are used in drills, [milling cutters](#), single point lathe tools, [broaches](#).

Cutting speed range - 30-50 m/min

Temperature - 650°C

Hardness – up to HRC 67

T-Type - Tungsten predominant type

M-Type - Molybdenum dominant type

Cemented carbide tool and cermet

The cemented carbide cutting tool is produced by powder metallurgy technique. It consists of tungsten, tantalum and titanium carbide with cobalt as a binder (when the binder is nickel or molybdenum, then it is called cermet). Cemented carbide tools are extremely hard; they can withstand very high-speed cutting operation. Carbide tool does not lose their hardness up to 1000° C. A high cobalt tool is used for a rough cut while low cobalt tool used for finishing operations.

Cutting speed range - 60-200m/min

Temperature - 1000°C

Hardness – up to HRC 90

Ceramics

Most common ceramic materials are aluminum oxide and silicon nitride. Powder of ceramic material Compacted in insert shape, then [sintered](#) at high temperature. Ceramic tools are chemically inert and possess resistance to corrosion. They have high compressive strength. They are stable up to temperature 1800°C. They are ten times faster than HSS. The friction between the tool face and chip are very low and possess low heat conductivity, usually no coolant is required. They provide the very excellent surface finish.

Cutting speed 300-600m/min

Temperature - 1200°C

Hardness – up to HRC 93

Cubic boron nitride (CBN)

It is the second hardest material after diamond. They are generally used in hand machines. They offer high resistance to abrasion and use as an abrasive in grinding wheels. Sharp edges are not recommended.

Speed 600-800m/min

Hardness - higher than HRC 95

Diamond

It is the hardest material known and it is also expensive. It possesses very high thermal conductivity and melting point. Diamond offers excellent abrasion resistance, low friction coefficient and low thermal expansion. It is used in machining very hard material such as carbides, nitrides, glass, etc. Diamond tools give a good surface finish and dimensional accuracy. They are not recommended for machining steel.

